FLATNESS CHECKED ON STALKS AT A CONSERVATIVE FAMILY OF POINTS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three same-chapter lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Stalks of flat modules are flat. Stacks Project, Modules on Sites, Lemma lemma-stalk-flat (carded locally as S11-sites-mod-stalk-flat). Supplies the forward implication.
- Stalks commute with tensor products. Stacks Project, Modules on Sites, Lemma lemma-tensor-product-pullback. Supplies the identity (\mathcal{F} \otimes \mathcal{G})_p = \mathcal{F}_p \otimes \mathcal{G}_p used in the converse.
- Exactness checked on stalks at a conservative family. Stacks Project, Modules on Sites, Lemma lemma-check-exactness-stalks. Supplies that exactness of a sequence of modules is detected at the stalks of a conservative family of points.

Source reference. Original lemma 05VB: sources/stacks-chow/tags.txt, source lines 7545–7545. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03EL: sources/stacks-chow/tags.txt, source lines 4460–4460. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05V3: sources/stacks-chow/tags.txt, source lines 7537–7537. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $\{p_i\}_{i \in I}$ be a conservative family of points of $\mathcal{C}$. Then $\mathcal{F}$ is flat if and only if $\mathcal{F}_{p_i}$ is a flat $\mathcal{O}_{p_i}$-module for all $i \in I$.*

Proof. By Lemma 05VB we see one of the implications. For the converse, use that $(\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G})_p = \mathcal{F}_p \otimes_{\mathcal{O}_p} \mathcal{G}_p$ by Lemma 03EL (as taking stalks at p is given by p^{-1}) and Lemma 05V3. □

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